

# Visualization Libraries Documentation

## Matplotlib & Seaborn

ShadowFox Data Science Internship

K. Ramya Reddy

### 1. Introduction:

**Data Visualization** is the graphical representation of data to understand patterns, trends, and relationships. It helps in making better decisions by presenting complex data in an easy-to-understand format.

In Python, there are many visualization libraries. Among them, **Matplotlib** and **Seaborn** are widely used.

**Matplotlib** is a powerful and flexible library used for creating various types of plots and charts.

**Seaborn** is built on top of Matplotlib and provides more attractive and statistical visualizations with simpler syntax.

### 2. Matplotlib:

Matplotlib is a Python library used for creating static, animated, and interactive visualizations. It provides full control over plot customization and is widely used in data analysis.

#### 2.1 Line Plot:

##### Description:

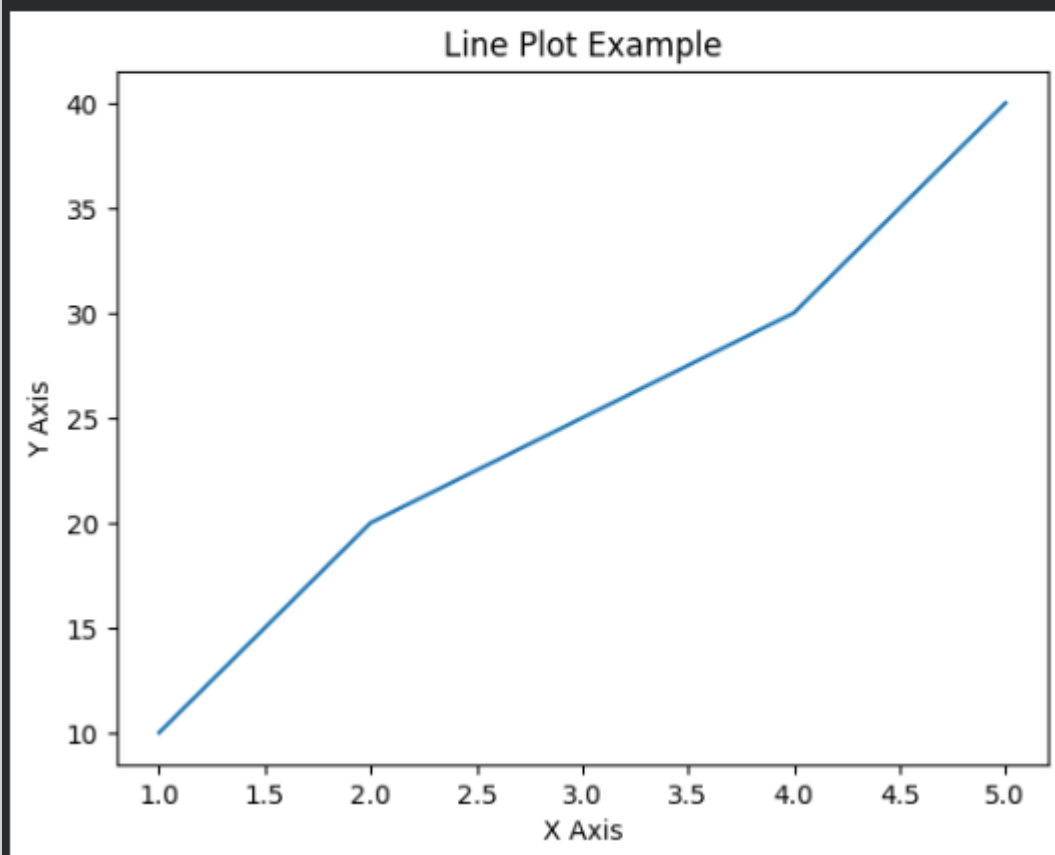
A line plot connects data points with a continuous line. It is used to show trends over time.

**Use Case:**

Used to display sales trends, temperature changes, stock prices, etc.

code:

```
import matplotlib.pyplot as plt
x = [1, 2, 3, 4, 5]
y = [10, 20, 25, 30, 40]
plt.plot(x, y)
plt.title("Line Plot Example")
plt.xlabel("X Axis")
plt.ylabel("Y Axis")
plt.show()
```



## 2.2 Bar Chart:

**Description:**

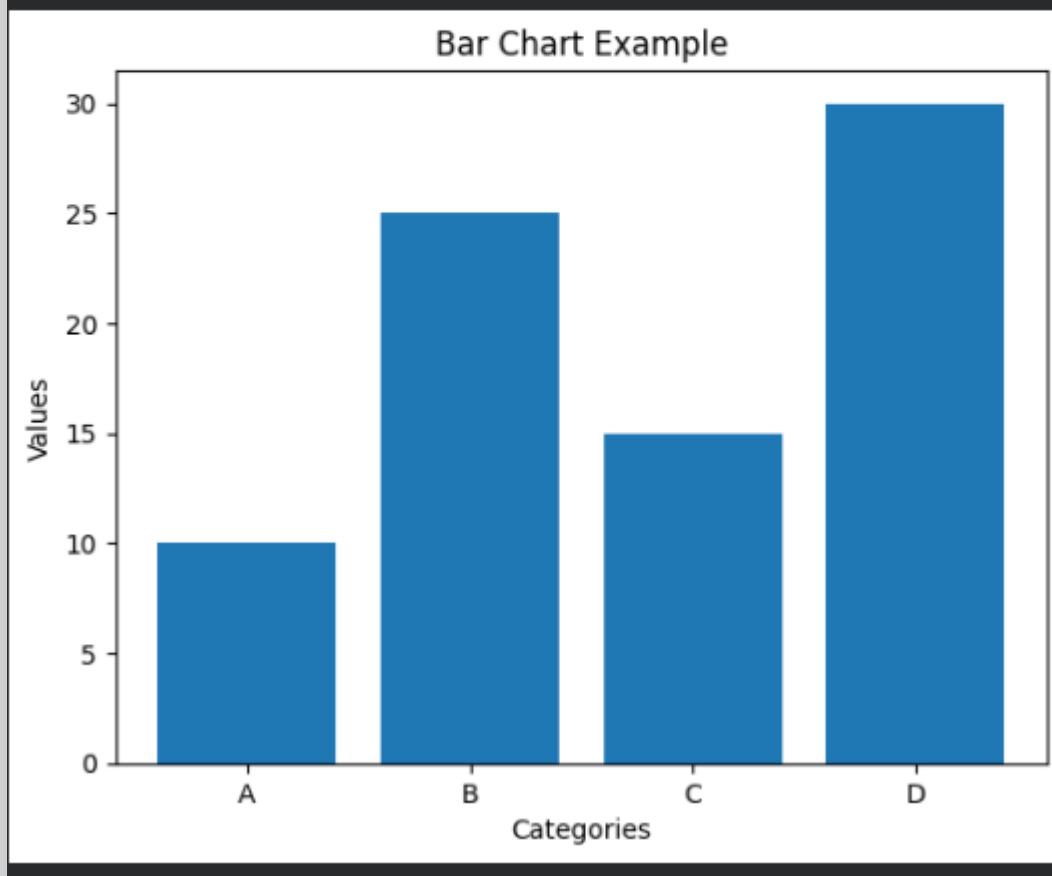
A bar chart represents data using rectangular bars. The height of the bar shows the value.

**Use Case:**

Used to compare categories like sales of products, student marks, etc.

Code:

```
import matplotlib.pyplot as plt
categories = ["A", "B", "C", "D"]
values = [10, 25, 15, 30]
plt.bar(categories, values)
plt.title("Bar Chart Example")
plt.xlabel("Categories")
plt.ylabel("Values")
plt.show()
```



## 2.3 Scatter Plot:

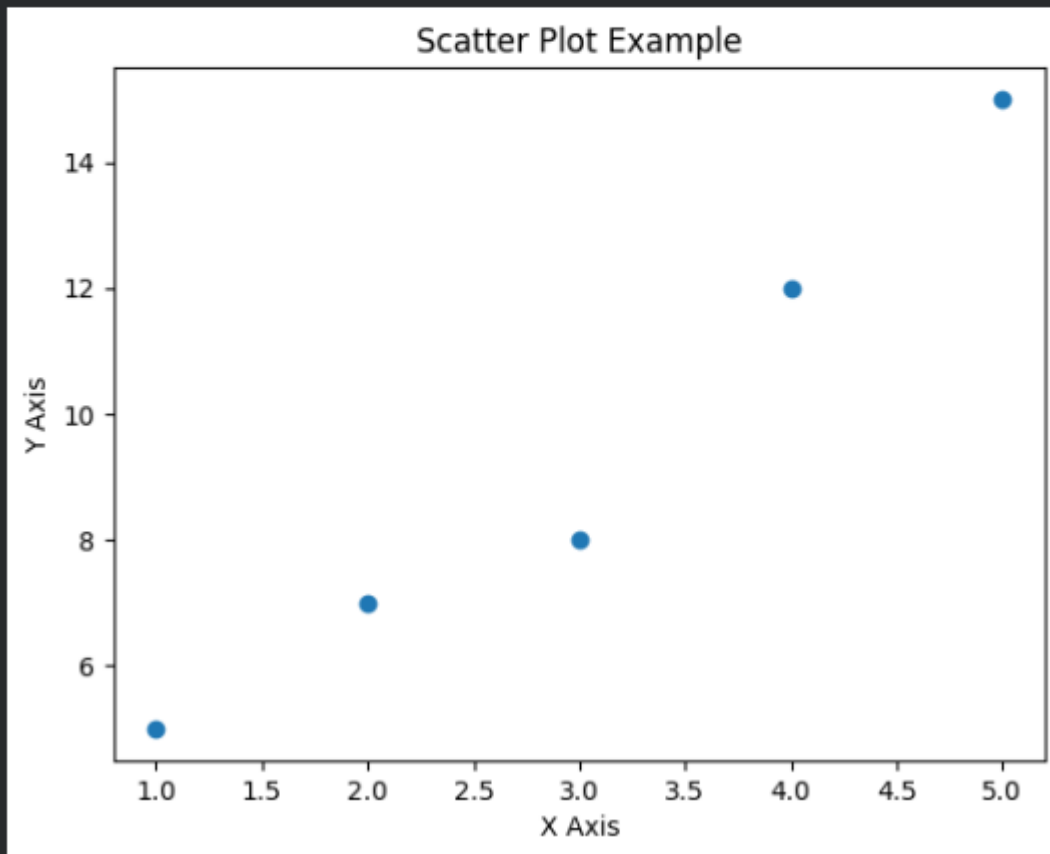
**Description:**

A scatter plot shows the relationship between two variables using dots.

**Use Case:**

Used to identify correlation between variables like height and weight.

```
import matplotlib.pyplot as plt
x = [1, 2, 3, 4, 5]
y = [5, 7, 8, 12, 15]
plt.scatter(x, y)
plt.title("Scatter Plot Example")
plt.xlabel("X Axis")
plt.ylabel("Y Axis")
plt.show()
```



## 2.4 Histogram

### Description:

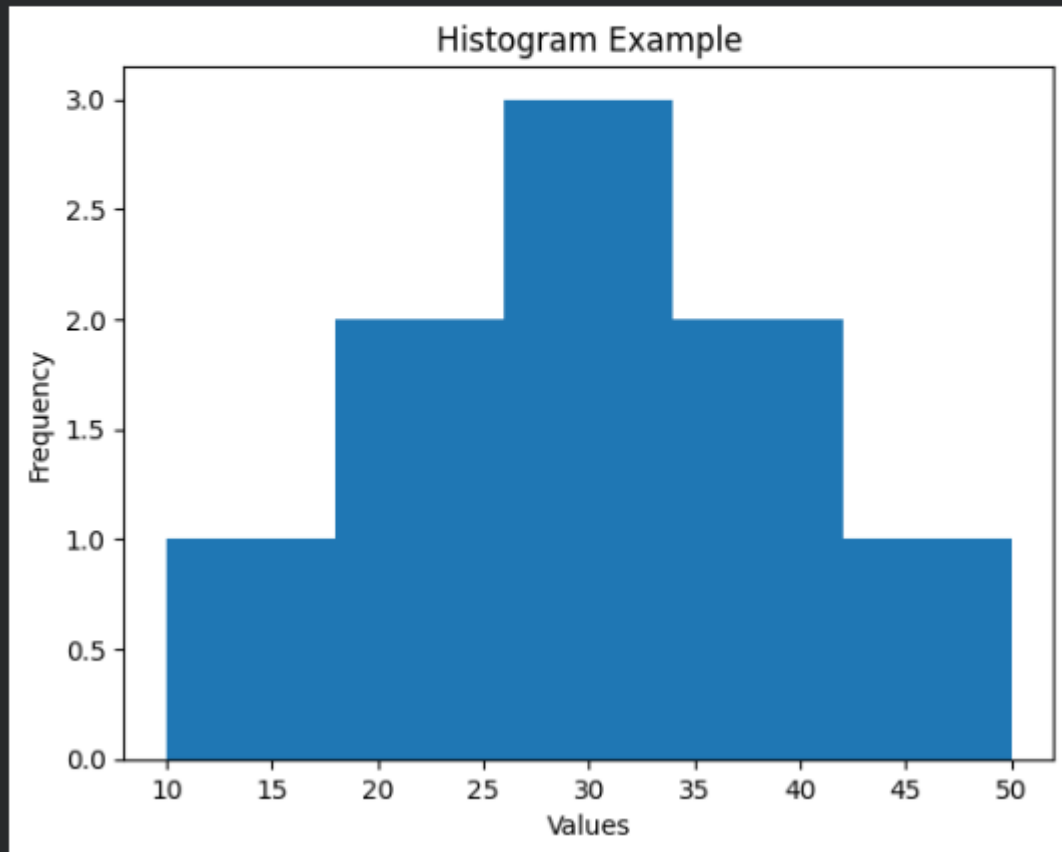
A histogram shows the distribution of numerical data by dividing it into bins.

### Use Case:

Used to analyze frequency distribution like exam scores.

Code:

```
import matplotlib.pyplot as plt
data = [10, 20, 20, 30, 30, 30, 40, 40, 50]
plt.hist(data, bins=5)
plt.title("Histogram Example")
plt.xlabel("Values")
plt.ylabel("Frequency")
plt.show()
```



## 2.5 Pie Chart

### Description:

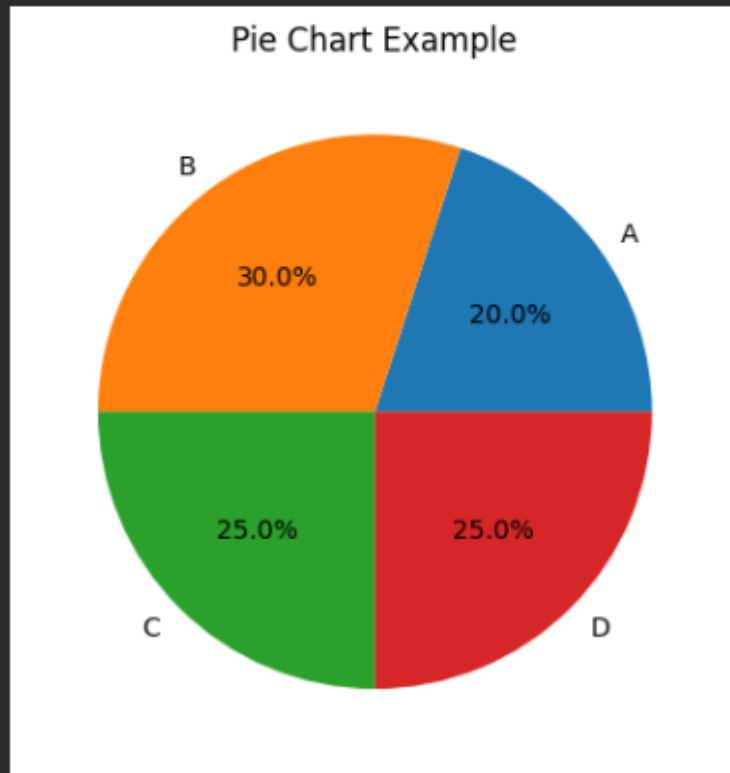
A pie chart represents data as slices of a circle based on percentage.

### Use Case:

Used to show proportion like market share or budget distribution.

Code:

```
import matplotlib.pyplot as plt
labels = ["A", "B", "C", "D"]
sizes = [20, 30, 25, 25]
plt.pie(sizes, labels=labels, autopct="%1.1f%%")
plt.title("Pie Chart Example")
plt.show()
```



### 3. Seaborn:

Seaborn is a Python data visualization library built on top of Matplotlib. It provides attractive and informative statistical graphics with less code. Seaborn works very well with Pandas DataFrames and has built-in themes and color palettes.

#### 3.1 Line Plot-

##### Description:

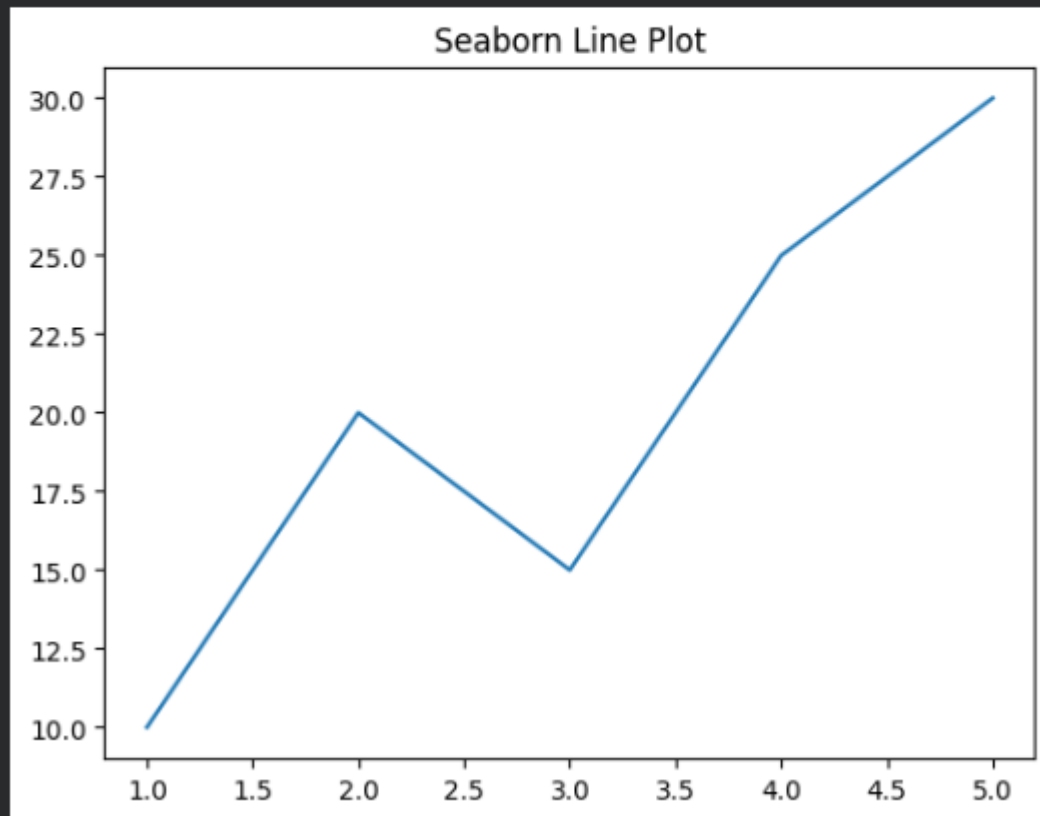
A line plot shows trends over time using lines.

##### Use Case:

Used to analyze trends like monthly sales or temperature changes.

Code:

```
import seaborn as sns
import matplotlib.pyplot as plt
x = [1, 2, 3, 4, 5]
y = [10, 20, 15, 25, 30]
sns.lineplot(x=x, y=y)
plt.title("Seaborn Line Plot")
plt.show()
```



### 3.2 Bar Plot-

#### Description:

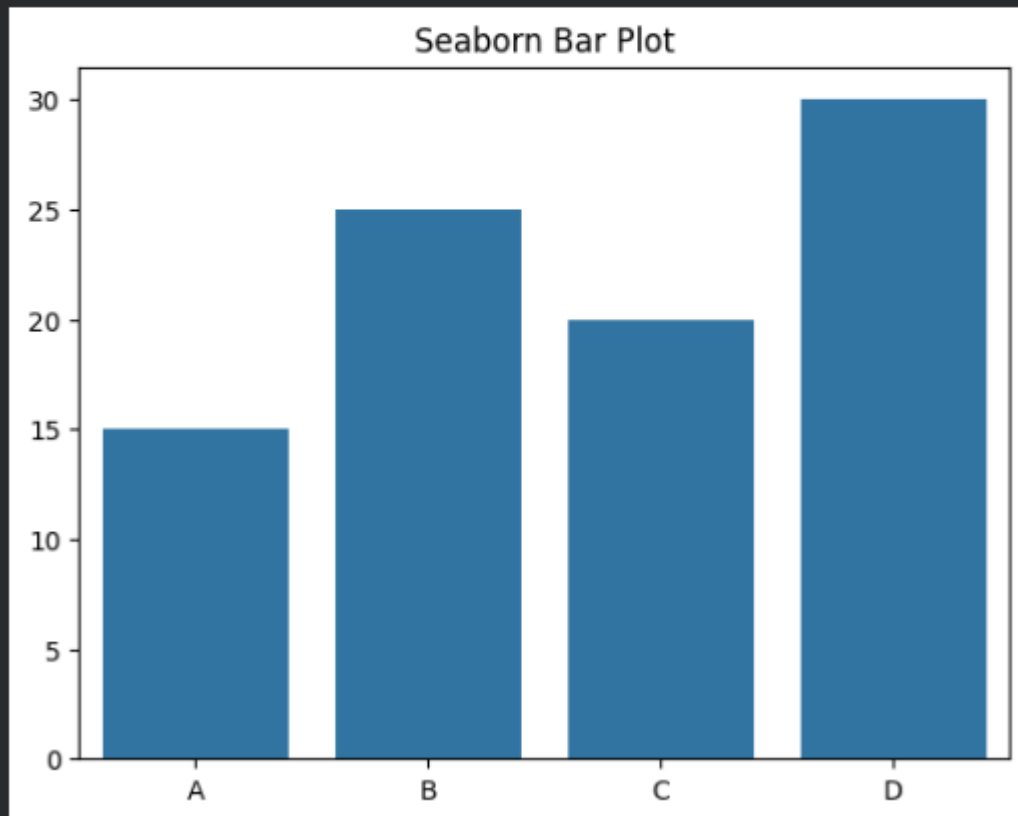
A bar plot compares different categories.

#### Use Case:

Used to compare performance of different groups.

Code:

```
import seaborn as sns
import matplotlib.pyplot as plt
categories = ["A", "B", "C", "D"]
values = [15, 25, 20, 30]
sns.barplot(x=categories, y=values)
plt.title("Seaborn Bar Plot")
plt.show()
```



### 3.3 Scatter Plot-

#### Description:

A scatter plot shows relationship between two variables.

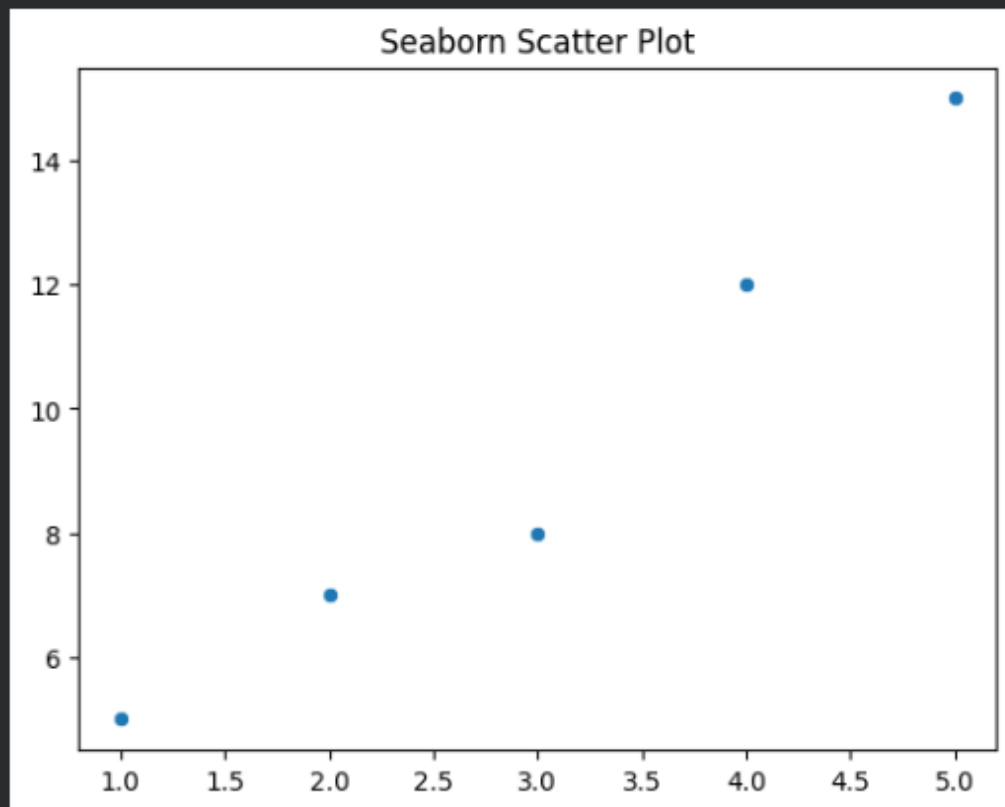
#### Use Case:

Used to check correlation between two numerical values.

Code:



```
import seaborn as sns
import matplotlib.pyplot as plt
x = [1, 2, 3, 4, 5]
y = [5, 7, 8, 12, 15]
sns.scatterplot(x=x, y=y)
plt.title("Seaborn Scatter Plot")
plt.show()
```



### 3.4 Box Plot-

#### Description:

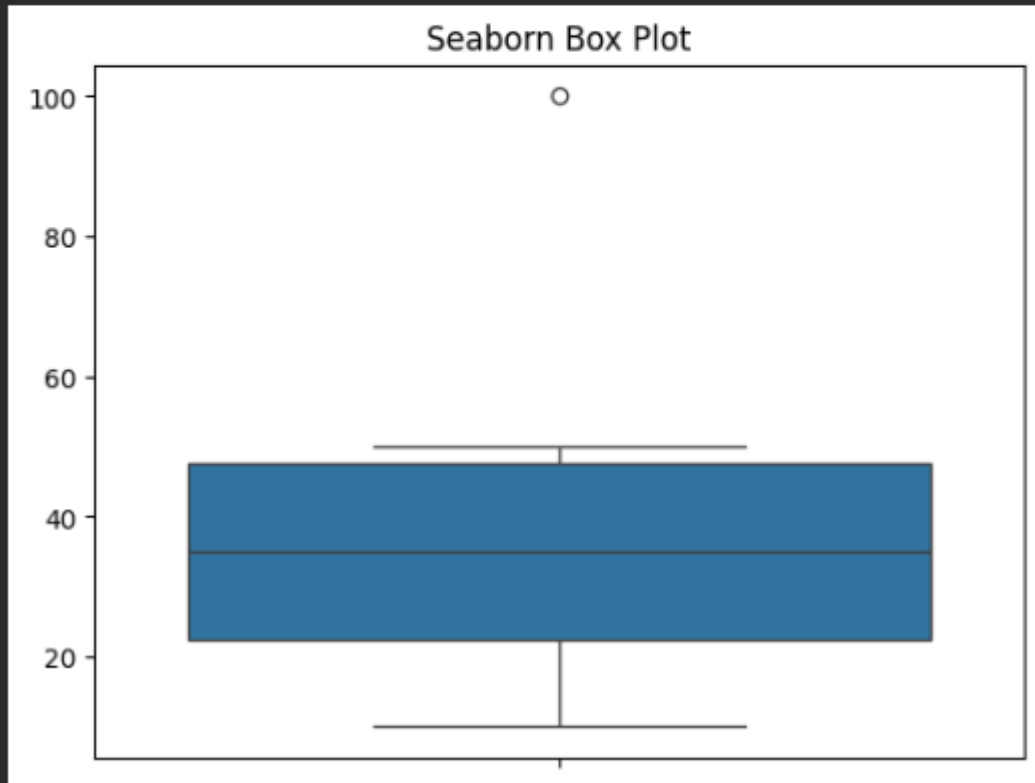
A box plot shows data distribution, quartiles, and outliers.

#### Use Case:

Used to detect outliers and compare distributions.

Code:

```
import seaborn as sns
import matplotlib.pyplot as plt
data = [10, 20, 30, 40, 50, 100]
sns.boxplot(y=data)
plt.title("Seaborn Box Plot")
plt.show()
```



### 3.5 Heatmap-

**Description:**

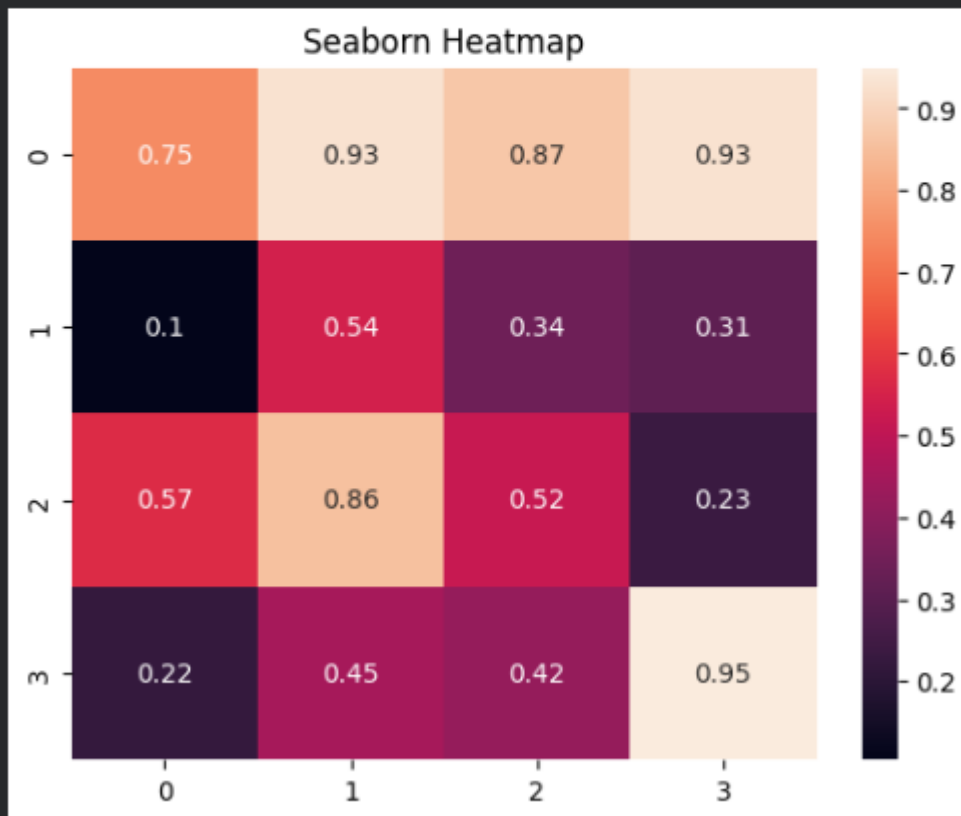
A heatmap shows data using colors. Higher values are darker or brighter.

**Use Case:**

Used to visualize correlation matrix.

Code:

```
import seaborn as sns
import matplotlib.pyplot as plt
import numpy as np
data = np.random.rand(4,4)
sns.heatmap(data, annot=True)
plt.title("Seaborn Heatmap")
plt.show()
```



## 4. Comparison Between Matplotlib and Seaborn:

Matplotlib and Seaborn are both powerful visualization libraries in Python. However, they differ in ease of use, customization, and purpose.

| Feature         | Matplotlib                    | Seaborn                   |
|-----------------|-------------------------------|---------------------------|
| Ease of Use     | Moderate (more code required) | Easy (less code required) |
| Customization   | Very High                     | Moderate                  |
| Built-in Styles | Basic                         | Attractive default themes |

| Feature           | Matplotlib                            | Seaborn                                |
|-------------------|---------------------------------------|----------------------------------------|
| Statistical Plots | Limited                               | Strong statistical support             |
| Integration       | Works with NumPy                      | Works well with Pandas                 |
| Best For          | Full control & detailed customization | Quick and beautiful statistical graphs |

Matplotlib provides complete control over every element of a plot, making it suitable for highly customized and complex visualizations. However, it requires more code and effort.

Seaborn simplifies visualization by providing built-in themes, color palettes, and statistical plotting functions. It is ideal for quick data analysis and visually appealing graphs with minimal coding effort.

## 5. Conclusion:

Both Matplotlib and Seaborn are essential libraries for data visualization in Python. Matplotlib is best suited for users who need detailed customization and flexibility. Seaborn is ideal for statistical analysis and creating attractive visualizations quickly. Depending on the project requirements, both libraries can be used together to achieve effective data visualization.